

Production runbook (FundFlow SaaS)

Concise operator checklist after deploy or tenant onboarding.

Deploy / release

1. Pull release tag on the app server.
2. `composer install --no-dev --optimize-autoloader`
3. `php artisan migrate --force` (central database)
4. `php artisan tenants:migrate --no-interaction`
5. `npm ci && npm run build` (Filament tenant/member/central themes)
6. `php artisan optimize:clear` then `php artisan config:cache / route:cache` if you use caching in production
7. Restart **queue workers**, **Reverb**, and PHP-FPM (or Sail/containers) so code and config reload:

```
sudo supervisorctl restart fundflow-reverb
# sudo supervisorctl restart fundflow-queue:* # if you use a queue worker unit
sudo systemctl reload php8.4-fpm
```

Reverb & real-time notifications

The admin/member **notification bell** can push over WebSockets when `BROADCAST_CONNECTION=reverb`. Laravel sends events to Reverb at `127.0.0.1:8080` (`REVERB_SERVER_*` in `.env`). Browsers connect via `wss://<CENTRAL_DOMAIN>` (`REVERB_HOST / VITE_REVERB_*`).

Prerequisites

Requirement	Notes
Redis	<code>CACHE_STORE=redis</code> needs <code>php8.4-redis + redis-server</code> (Reverb start uses cache; Stancl tagged tenant cache).
<code>.env</code>	<code>BROADCAST_CONNECTION=reverb</code> , Reverb app keys, <code>REVERB_SERVER_HOST=127.0.0.1</code> , <code>REVERB_SERVER_PORT=8080</code> , <code>FILAMENT_DATABASE_NOTIFICATIONS_POLLING=10s</code> (fallback polling).
Frontend build	<code>npm run build</code> so <code>VITE_REVERB_*</code> are in the manifest.
Nginx	Proxy WebSocket upgrades from public HTTPS to <code>127.0.0.1:8080</code> (see below).

If Reverb is not running, the bell still updates every **10s** via Livewire polling, but logs may show **Pusher error: cURL error 7 ... Couldn't connect to 127.0.0.1 port 8080**.

Supervisor (Reverb)

Operator guide: [docs/reverb-supervisor.md](#) — install Supervisor, Redis/phpredis, install fundflow-reverb, command cheat sheet, troubleshooting.

Use the unit file in the repo (do **not** run a second manual `reverb:start` in SSH — that causes Address already in use on 8080):

```
sudo apt install -y supervisor redis-server php8.4-redis # one-time
sudo cp /var/www/fundflow-saas/deploy/supervisor/fundflow-reverb.conf /etc/supervisor/conf.d/
sudo supervisorctl reread && sudo supervisorctl update && sudo supervisorctl start fundflow-reverb
sudo supervisorctl status fundflow-reverb
```

Logs: `storage/logs/reverb.log` — `tail -f storage/logs/reverb.log`

After code or .env changes: `php artisan config:clear` then `sudo supervisorctl restart fundflow-reverb`

Graceful signal to running server: `php artisan reverb:restart`

Nginx WebSocket proxy (example)

Add a location on the same `server` block that serves the app (adjust `server_name` / TLS paths). Reverb must stay bound to localhost only:

```
location /app/ {
    proxy_http_version 1.1;
    proxy_set_header Host $http_host;
    proxy_set_header Scheme $scheme;
    proxy_set_header SERVER_PORT $server_port;
    proxy_set_header REMOTE_ADDR $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "Upgrade";
    proxy_pass http://127.0.0.1:8080;
}
```

Reload nginx: `sudo nginx -t && sudo systemctl reload nginx`

Filament Echo uses `/broadcasting/auth` on the main app for private channels; the `/app/` path is for the Reverb/Pusher protocol.

Smoke checks

```
ss -tlnp | grep 8080 # should show php (reverb)
redis-cli ping # PONG
sudo supervisorctl status fundflow-reverb
```

In the browser (admin or member panel): DevTools → Network → WS — expect a connection to your domain without errors when Reverb and nginx are correct.

Troubleshooting

Log / symptom	Cause	Action
Address already in use on 8080	Second <code>reverb:start</code> or stray process	<code>sudo supervisorctl status fundflow-reverb</code> ; stop manual duplicates; use <code>sudo supervisorctl restart fundflow-reverb</code> only.
Couldn't connect to 127.0.0.1 port 8080	Reverb not running	<code>sudo supervisorctl start fundflow-reverb</code>
Class "Redis" not found	Missing PHP redis extension	<code>sudo apt install php8.4-redis redis-server</code>
Echo fails in browser, polling works	nginx WS proxy or <code>VITE_REVERB_*</code> / build	Fix proxy; <code>npm run build</code> ; hard refresh

To disable broadcasting temporarily (polling only): set `BROADCAST_CONNECTION=log`, then `php artisan config:clear`.

More detail: [docs/filament-notification-bell-and-reverb.md](#).

Scheduled fund jobs

Ensure the host cron runs Laravel's scheduler every minute.

Recommended (system cron file):

```
cd /var/www/fundflow-saas
touch storage/logs/scheduler.log
sudo chown www-data:www-data storage/logs/scheduler.log
sudo cp deploy/cron/fundflow-scheduler /etc/cron.d/fundflow-scheduler
sudo chmod 644 /etc/cron.d/fundflow-scheduler
sudo chown root:root /etc/cron.d/fundflow-scheduler
```

The template at `deploy/cron/fundflow-scheduler` runs:

```
* * * * * www-data cd /var/www/fundflow-saas && php artisan schedule:run >> storage/logs/scheduler.log
```

Manual crontab alternative (`sudo crontab -u www-data -e`):

```
* * * * * cd /var/www/fundflow-saas && php artisan schedule:run >> /dev/null 2>&1
```

Verify:

```
php artisan schedule:list
tail -f /var/www/fundflow-saas/storage/logs/scheduler.log
```

Review **System** → **Jobs & commands** in the tenant admin panel for the catalog, last run times, and manual runs (respects `BatchPostingGate`).

Tenant admin URLs (typical)

Task	Path
Dashboard	<code>/admin</code>
Contribution cycles	Use nav Fund Management → Contribution cycles (Filament-discovered slug)
Migration workflow	<code>/admin/migration-workflow</code>

Task	Path
Reconciliation	/admin/reconciliation-exceptions
Jobs	/admin/jobs

After navigation changes, run `php artisan optimize:clear` if menus look stale.

Migration onboarding (legacy members)

1. Open **Fund Management** → **Migration workflow** (or **Members** → **Migration workflow**).
2. **Not started** tab → **Begin migration** (cutoff date) or use **Migration** → **Generate migration stubs** on the member profile.
3. On the member profile: **Migration cycles** tab — classify stubs, lump-sum / instalment / OB offset settlements.
4. **Migration** → **Post opening balances**, then **Clear for active operation** (or grant partial clearance when escalated cycles remain open).

Smoke checks

```
php artisan test --compact tests/Feature/Tenant/ComplianceLayerTest.php tests/Feature/Tenant/MigrationWorkflowTest.php
```

Verify nightly reconciliation and contribution window commands appear healthy on **Jobs & commands**.